★TOPIC:- HARDWARE

Module_1-Day_4:

❖ Tasks:

1. **Open your own laptop or desktop (or use an online PC hardware simulator if you don't have access to a real system) and identify the GPU model installed. Write down the GPU's brand, model number, and whether it is integrated or dedicated.**

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❖ GPU Identification:

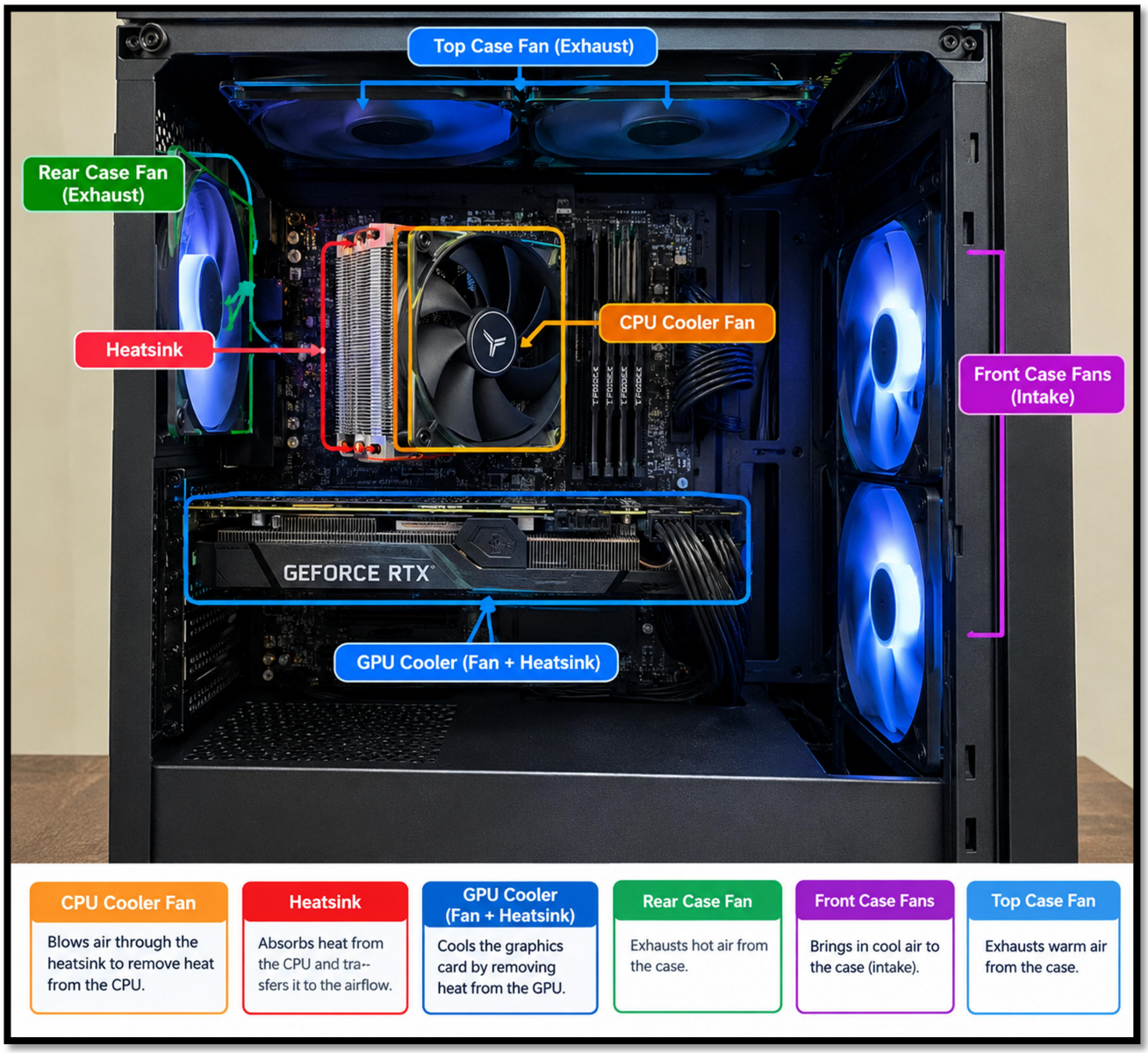
Detail	Answer
GPU Brand	Intel
GPU Model	Intel UHD Graphics 620 (likely; verify on your laptop)
GPU Type	Integrated GPU
Location	Built into the processor

2. **Take a clear photo or screenshot of the cooling system (fan, heatsink, or liquid cooling setup) inside your PC or from a PC hardware simulator. Label the main cooling components and briefly describe their function.**

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❖ PC Cooling System – Components and Functions:

- ★ **CPU Cooler Fan:** Removes heat from the CPU by blowing air over the heatsink.
- ★ **Heatsink:** Absorbs heat from the CPU and transfers it to the surrounding air.
- ★ **GPU Cooler:** Keeps the graphics card cool during operation.
- ★ **Rear Case Fan (Exhaust):** Pushes hot air out of the PC case.
- ★ **Front Case Fans (Intake):** Bring cool air into the PC case.
- ★ **Top Case Fans (Exhaust):** Remove hot air from the top of the PC case.



CPU Cooler Fan
Blows air through the heatsink to remove heat from the CPU.

Heatsink
Absorbs heat from the CPU and tra-- sfers it to the airflow.

GPU Cooler (Fan + Heatsink)
Cools the graphics card by removing heat from the GPU.

Rear Case Fan
Exhausts hot air from the case.

Front Case Fans
Brings in cool air to the case (intake).

Top Case Fan
Exhausts warm air from the case.

3. **Create a comparison table listing at least 3 differences between air cooling and liquid cooling systems for GPUs, including factors like cost, efficiency, and maintenance.**

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❖ **Difference Between Air Cooling and Liquid Cooling for GPUs:**

Point	Air Cooling	Liquid Cooling
Cost	Cheaper and budget-friendly.	More expensive to buy and install.
Efficiency	Good cooling for normal gaming and daily use.	Can provide better cooling for high-performance GPUs.
Maintenance	Easy to maintain; mainly requires dust cleaning.	Requires more care; liquid leaks can be a risk.
Noise	Fans can become noisy at high speeds.	Can be quieter, but the pump and radiator fans also make noise.
Installation	Easy to install and use.	More complex to install, especially custom liquid cooling.
Example	Standard GPU with a heatsink and fans.	GPU with a liquid-cooling block, pump, and radiator.

4. **Imagine you are building a gaming PC for streaming IPL matches and playing graphics-heavy games. Choose between air cooling and liquid cooling for your GPU, and justify your choice in 3-4 sentences based on your research.**

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★ I would choose **air cooling** because it is affordable, easy to install, and requires less maintenance. It is sufficient for streaming IPL matches and playing most games when properly matched to the GPU.

★ I would choose **liquid cooling** because it can provide effective heat removal for high-performance systems during long gaming sessions. Although it is more expensive and requires more care, it can help maintain stable temperatures under heavy workloads.